

CUSTOMER CHURN PREDICTION USING MACHINE LEARNING TECHNIQUES

CLASSIFICATION VS ENSEMBLE LEARNING MODELS



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TOOLS USED

- Python
- Pandas
- Scikit-Learn
- XGBoost

1 INTRODUCTION



Customer churn occurs when customers stop using a company's services. Predicting churn helps businesses take proactive actions to retain valuable customers and reduce revenue loss.

2 DATASET OVERVIEW



7043
RECORDS

21
FEATURES



Customer Details



Services Used



Billing Information



Usage Metrics



Churn Label

3 MACHINE LEARNING WORKFLOW



Data Collection



Data Preprocessing



Exploratory Data Analysis



Feature Engineering



Model Training



Model Evaluation



Best Model Selection

4 MODELS USED



LOGISTIC REGRESSION

Linear model for classification



DECISION TREE

Tree-based model with decision rules



RANDOM FOREST

Ensemble of decision trees for robust predictions



XGBOOST

Gradient boosting for high performance

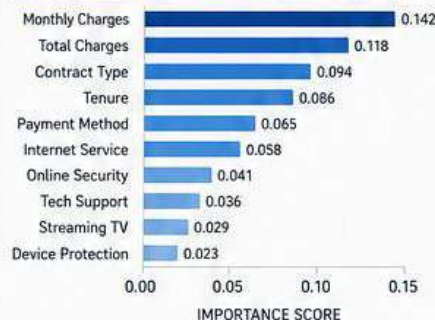
5 ACCURACY COMPARISON



6 CONFUSION MATRIX (XGBOOST)



7 FEATURE IMPORTANCE (XGBOOST)



8 FINAL RESULT



XGBOOST

IS THE BEST PERFORMING MODEL

94.91%
ACCURACY

High accuracy and better generalization make XGBoost the optimal choice for customer churn prediction.



Predict. Prevent. Retain.
Building Intelligent Solutions for a Better Tomorrow.

